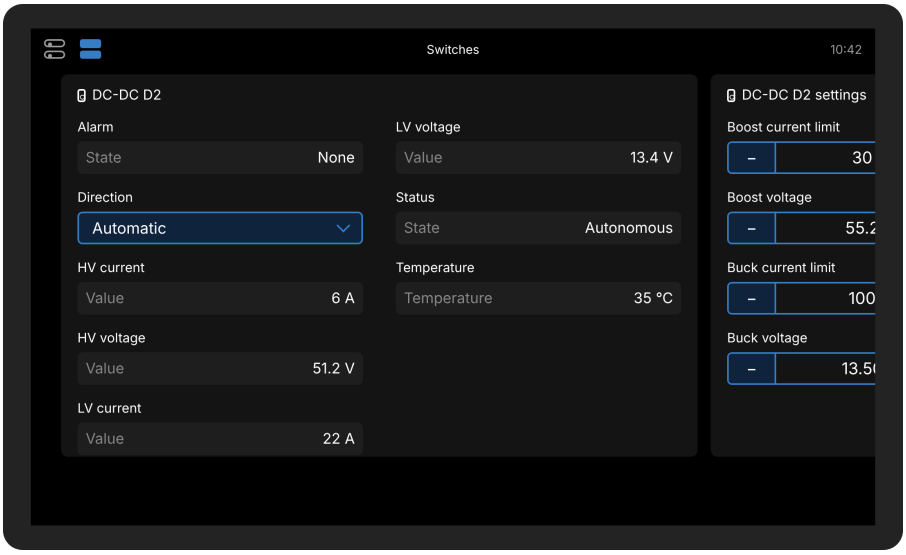
INSTALLATION GUIDE · CONTROL EDITION V1.4

Nomadic D2 DC-DC converter on your Victron Cerbo GX

Make the cable, set up the Cerbo, and install the driver. When you're done, the D2's readings and controls appear on the Cerbo screen, in the VRM app and on a web page.

> **This is the control edition, for advanced users.** It can command the D2. If you only want to see what the D2 is doing, use the monitor-only edition instead (`github.com/CKOCI13/CKOCI13-cerbo-nomadic-d2-monitor`), which never sends the D2 anything.



What you'll see on the Cerbo screen when you're done (drawn from Victron's own screen design, so the real screen may differ slightly).

You'll be able to

- See LV/HV voltage and current, temperature, status and alarms
- Run the D2 in Automatic within your current limits, or choose Off, Buck (48 V to 12 V) or Boost (12 V to 48 V) by hand
- Change the voltage targets and current limits

Good to know

- It takes about 1 hour, including making the cable
- In Automatic (the default) the Cerbo makes the D2's own buck / boost / standby choices and keeps it within your current limits. Nobody has to choose Buck or Boost
- You don't need to type any commands. The **Cerbo Setup** helper does it for you

STEP 1

STEP 2
Make the cable

STEP 3

STEP 4

STEP 5
Install the driver

STEP 6
Use it

Get what you need

Set up the Cerbo

Connect everything

1 Get what you need

Gather everything before you start.

For the cable

- A standard Ethernet patch cable (Cat5e or Cat6), long enough to reach from the Cerbo to the D2
- Two **3-way lever wire connectors**, for example WAGO 221-413. They take 24–12 AWG wire: solid-core Cat5e/Cat6 (about 24 AWG) fits; very thin stranded patch-cable wire may not clamp well, so use solid-core cable, or put a small ferrule on each end
- Wire cutters and a wire stripper
- Heat-shrink tubing or electrical tape
- A multimeter with a continuity (beep) setting and an Ohms setting
- A VE.Can terminator: the small RJ45 plug that came in the Cerbo box
- A **120 Ω resistor**, ¼ W metal film (1 %), for the D2 end. Measure it before use: 115–125 Ω

For the software

- A Windows PC on the same network (Wi-Fi or Ethernet) as the Cerbo
- The project folder downloaded from GitHub, which contains **Cerbo Setup.bat** and the **dbus-nomadic-dcdc** folder
- The Cerbo running Venus OS **v3.80 or newer**. Step 3 shows you how to check

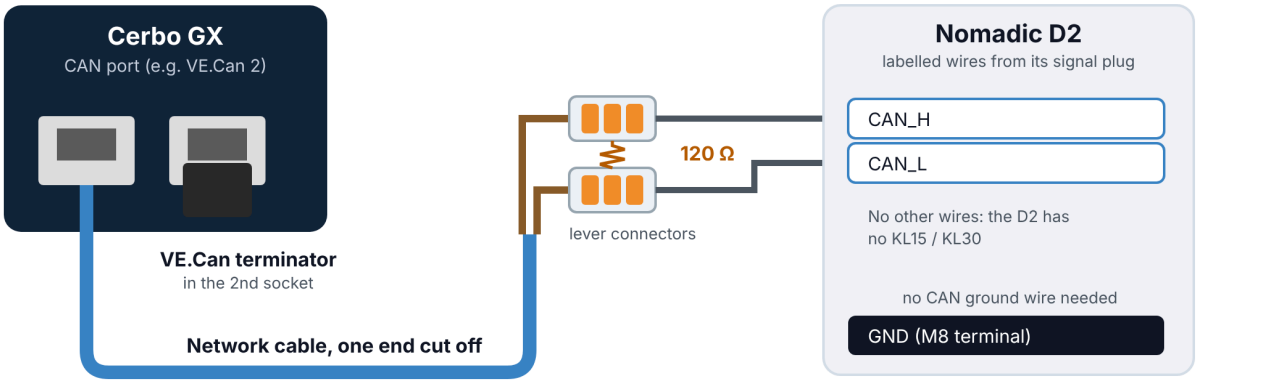
Safety

Before you work on the D2's wiring, open the fuses or breakers on **both** its LV+ (12 V) and HV+ (48 V) leads and follow the D2 manual. The D2 has no KL30/KL15 supply to switch off: it powers itself from whichever power terminal is live. The CAN cable only carries data, but it ends next to the D2's 48 V and 12 V power terminals.

2 Make the cable

The Cerbo uses **RJ45 sockets**, the same plug as network cables. Take a ready-made network cable, keep the plug on one end for the Cerbo, and cut off the other end. You only need **2 of its 8 wires**: the brown pair. The D2 already has labelled wires coming out of its signal plug (CAN_H and CAN_L). You join the network cable to its CAN_H and CAN_L wires with **two lever wire connectors**. The **120 Ω resistor** goes between the two connectors. **No ground wire** is needed.

The whole connection

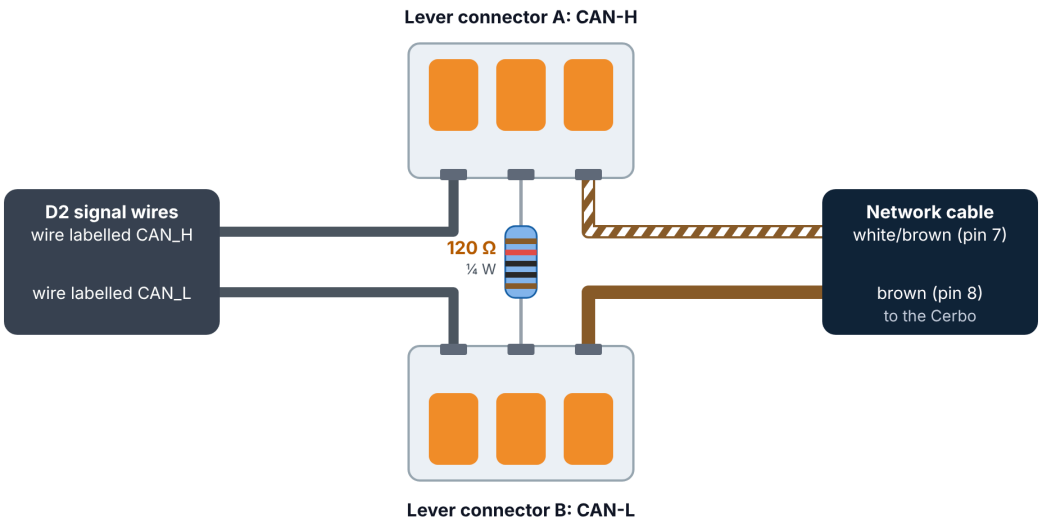


white/brown = CAN_H, brown = CAN_L. White/green and the other wires: not used.

The D2 powers its controller from its LV+ or HV+ terminal.

Blue: the network cable you make. The two lever connectors join it to the D2's CAN_H and CAN_L wires, with the required 120 Ω resistor between them. The D2 has no KL30/KL15 wires: it powers itself from its LV+/HV+ terminals.

The two lever connectors

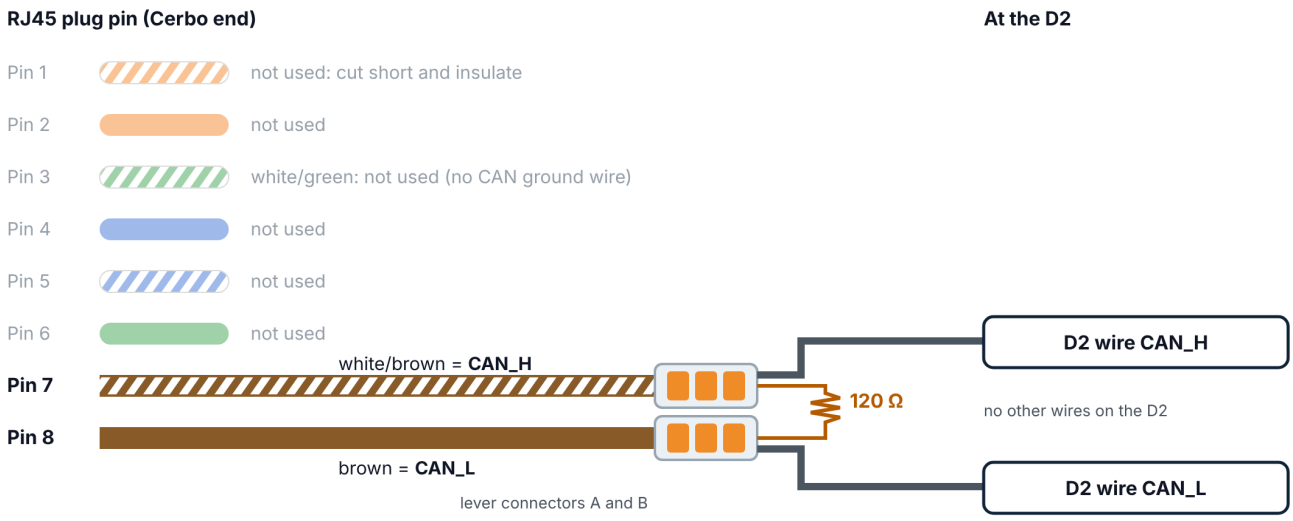


White/green (pin 3) and the other five wires: **not used**, cut short and insulate each one. No ground wire.

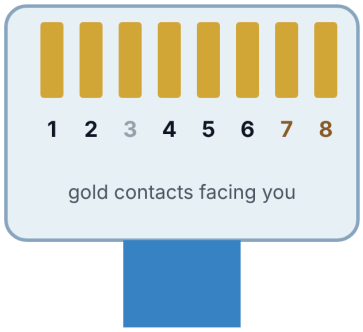
Strip each wire 11 mm (the gauge on the connector's side), push it fully in, close the lever. Heat-shrink both resistor leads.

Each connector joins three things: one D2 wire, one network-cable wire and one leg of the resistor. The resistor bridges the two connectors, so it sits across CAN_H and CAN_L at the D2.

Which wire goes where

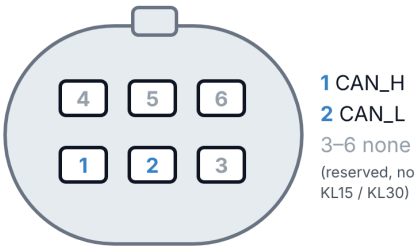


Pins 7 and 8 are the **brown pair** (twisted together), which suits CAN well. Everything else, including white/green (pin 3), is not used.



cable pointing down, clip at the back

RJ45 plug (Cerbo end): only pins 7 and 8 are used.



layout as drawn in the D2 manual (facing the D2's connector)

D2 signal plug (TE 1-967616-1), for reference only: its wires come out labelled, so there's no need to open it.

Wire	RJ45 pin (Cerbo)	Goes to
white/brown	7	**Lever connector A**, with the D2's wire labelled **CAN_H**
brown	8	**Lever connector B**, with the D2's wire labelled **CAN_L**
120 Ω resistor	–	One leg in connector A, the other in connector B
The other 6 wires (pins 1–6, including white/green)		Not used. Cut them short and insulate each one. No ground wire

Sources: Victron's *VE.Can to CAN-bus BMS cables manual* (Cerbo side: pin 7 CAN-H, pin 8 CAN-L) and the *D2 Bidirectional DC-DC Converter Manual*, Rev D.

Make it

2a **Cut off one plug.** Keep the other plug for the Cerbo. Strip about 5 cm (2 in) of the grey outer jacket.

2b **Check the wires with a multimeter** on the continuity (beep) setting. Touch one probe to a gold contact on the plug and the other to a wire:

- Pin 7 should beep with **white/brown**
- Pin 8 should beep with **brown**

2c **Insulate the 6 wires you don't need**, including white/green. Cut them short and cover **each one** separately with heat-shrink or tape, so none can touch anything.

2d **Join the wires with the two lever connectors**, with the D2 switched off. First, with everything off, measure across the D2's CAN_H and CAN_L wires: if it reads about 120 Ω the D2 has a terminator built in, so leave the resistor out. Strip each wire **11 mm** (the strip gauge on the connector's side), lift the lever, push the wire fully in, and close the lever.

- **Connector A (CAN-H):** the D2's wire labelled **CAN_H**, the **white/brown** wire, and one leg of the resistor
- **Connector B (CAN-L):** the D2's wire labelled **CAN_L**, the **brown** wire, and the other leg of the resistor

Slide heat-shrink over both resistor legs so they can't touch each other or anything metal. Keep the brown pair twisted up to a few centimetres from the connectors. Tie the connectors and cable to something solid, or put them in a small box, so road vibration can't pull on them.

2e **No ground wire.** CAN works on the difference between CAN_H and CAN_L. The Cerbo and the D2 already share the van's negative through their power wiring, so a separate CAN ground isn't needed. Leaving it off also avoids a thin wire running in parallel with the main negative cables (the Cerbo's VE.Can 2 port is not isolated). Once everything runs, step 5c includes a quick check of the ground difference.

> Never connect any of the network-cable wires to the D2's HV+ or LV+ terminals. They only join the D2's labelled CAN_H and CAN_L wires.

2f **No other wires.** Unlike the D1, the D2 has no KL30 or KL15. It powers itself from whichever of its LV+ and HV+ terminals has power, and it talks on CAN whenever it is powered. If you are replacing a D1, insulate its old KL15 and KL30 wires.

2g **Check the resistor.** With the RJ45 plug not yet in the Cerbo, measure between pins 7 and 8 of the RJ45 plug (or the white/brown and brown wires). It should read about **120 Ω**. If it reads open, a wire isn't fully in its lever connector.

2h **Final check:** plug the RJ45 into the Cerbo port you'll use, with the VE.Can terminator in that port's other socket. With everything switched off, measure between white/brown and brown again:

You measure	Meaning
About 60 Ω	Correct: both ends are terminated
About 120 Ω	The VE.Can terminator isn't in the Cerbo port's other socket, or the cable is in a different port
Very high or "OL"	The resistor at the D2 isn't connected, or a wire isn't fully in its lever connector. Re-check steps 2d and 2g

3 Set up the Cerbo

Do this on the Cerbo's touchscreen, or remotely with Remote Console in VRM. Every item below is found from the Settings menu.

3a **Check the firmware version.** Go to **Settings › General › Firmware**. It needs to be **v3.80 or newer**. If it's older, use **Online updates** on the same page to update first.
Then go to **Settings › General › Modification checks** and make sure **modifications are enabled** (that's the default). If they are off, Venus OS switches off the start-up script and the driver disappears after the next restart.

3b **Write down the Cerbo's IP address.** Go to **Settings › Connectivity › Ethernet** (or Wi-Fi) and read the IP address.
IP address: _____

3c **Choose the CAN port and speed.** Use the Cerbo's second CAN port, with nothing else plugged into it. The labels next to the ports tell you which Cerbo you have:

- **Cerbo GX MK2** (ports labeled **VE.Can 1** and **VE.Can 2**): go to **Settings › Connectivity › VE.Can port 2** (some firmware still calls it **BMS-Can port**) and set **CAN-bus profile** to **VE.Can (250 kbit/s)**. That is the default. The D2 talks at either 250 or 500 kbit/s, so start with 250. Step 5c tells you if the D2 is talking at the other speed. If it is, change the profile to **CAN-bus BMS LV (500 kbit/s)**. Older firmware calls it **CAN-bus BMS (500 kbit/s)**.
  Victron's own messages use a different kind of CAN address from the D2's, so they don't get in each other's way. Other guides may mention a profile called *Up, but no services*. The Cerbo screen no longer offers it, and you don't need it.
- **Original Cerbo GX** (second port labeled **BMS-Can**): go to **Settings › Connectivity › BMS-Can port**. This port only runs at 500 kbit/s, so leave it on **CAN-bus BMS (500 kbit/s)**. It works only if the D2 talks at 500 kbit/s. If step 5c reports bus errors on it, the D2 is at 250 kbit/s: use the VE.Can port (if nothing else is on it) with **VE.Can (250 kbit/s)** instead.

3d **Let your PC connect to the Cerbo** (this switches on SSH):

1. Go to **Settings › General › Access & Security › Access level** and choose **User & Installer**. The password is `ZZZ`.
2. Still on the Access & Security page, drag the list down with your finger and hold it for about 5 seconds (on Remote Console with a keyboard: hold the right-arrow key). The access level changes to **Superuser**.
3. Tap **Root password** and choose a password (at least 8 characters).
   Root password: _____
4. Switch **Enable SSH on LAN** to on.

3e **Optional, to change settings from VRM:** go to **Settings › VRM › VRM Portal access level** and set it to **Full**.

4 Connect everything

4a Plug the cable's RJ45 plug into the CAN port you chose in 3c, with the VE.Can terminator in the port's other socket (as in step 2h).

4b Plug the D2's signal plug back in and close the fuses or breakers on its LV+ and HV+ leads.

> Nothing changes for the D2 yet. The Cerbo only listens until you finish Step 5. After installing, it starts in **Automatic with limits**: it makes the D2's own buck / boost / standby choices and sends your current limits. To install it so that it only watches, set `automatic = hands_off` in step 5d (the Safe Test Plan does this for its first stages).

5 Install the driver with Cerbo Setup

5a **Open the helper.** In the project folder, double-click `Cerbo Setup.bat`. If Windows shows "Windows protected your PC", click **More info** and then **Run anyway**.

5b **Type the Cerbo's IP address** from step 3b and press **Enter**. The helper remembers it for next time, and then shows its menu:

```
  ============================================================
    Nomadic DC-DC converter  -  Cerbo GX setup
    Cerbo address: 192.168.1.50
  ============================================================

    1  Edit the settings (config.ini)
    2  Find which Cerbo port the converter is on    (listen only)
    3  Watch live readings from the converter       (listen only)
    4  Install or update the driver on the Cerbo
    5  Show the driver log

    U  Make a USB-stick installer (no network needed)

    9  Try it with a pretend converter (simulation - nothing is sent)

    6  Change the Cerbo address
    7  Fix "REMOTE HOST IDENTIFICATION HAS CHANGED"
    8  Uninstall the driver from the Cerbo
    0  Exit

  Type a number and press Enter: _
```

> **Want to see it first?** Option **9** runs everything with a pretend D2 and sends nothing on CAN (see the Safe Test Plan, `docs/safe-test-plan.pdf`).

> **The password prompt.** Every option that talks to the Cerbo asks for the root password from step 3d. **Nothing appears on screen while you type it.** That's normal: type it and press Enter.

5c **Type `2`: find the converter.** The helper copies the driver to the Cerbo and listens on every CAN port for 3 seconds. You should see something like this:

```
Venus OS v3.80 OK
Listening 3 s on each CAN port (nothing is transmitted)...
  can0    250 kbit/s  error-active  nothing from the converter
  can1    250 kbit/s  error-active  CONVERTER FOUND (391 frames)
The converter is on can1. config.ini has interface = auto, so the driver finds it there by itself.
Good.
```

Port names differ between Cerbo models (`can1`, or `vecan1` on the MK2). That's fine: with `interface = auto` the driver uses whichever port the D2 is on. If the D2 is set to the other speed, you'll see bus errors instead:

```
  can1    250 kbit/s  error-passive nothing decoded, but 288 bus errors: talking at a different
speed?
Something is talking on can1 at a different speed. On the Cerbo, set that port's CAN-bus
profile to "CAN-bus BMS LV (500 kbit/s)" and run this again.
```

Do what it says (step 3c), then type `2` again. If nothing is heard at all, see Troubleshooting, and check first that the D2 has power on its LV+ or HV+ terminal.

> **Ground check (no CAN ground wire).** Later, with the D2 working hard, measure between the Cerbo's negative terminal and the D2's GND terminal. A few tenths of a volt is normal. More than about 1 V points to a poor negative connection: fix that first. If the Cerbo ever can't hear the D2 although the wiring and the 60 Ω check are right, try a wire from RJ45 pin 3 (white/green) to the D2's GND as a test.

5d **Type `1` : edit the settings.** Notepad opens `config.ini`. Check these lines, then press Ctrl+S to save and close Notepad:

Line	What to put
`interface = auto`	Leave as auto: the driver finds the D2's CAN port by itself
`startup = auto`	Leave as auto: after every Cerbo restart it's in Automatic
`automatic = limited`	Leave as limited: in Automatic the Cerbo makes the D2's own buck / boost choices and keeps it within your current limits. (`hands_off` : Automatic sends nothing)
`boost_voltage = 55.2`	The 48 V target used when you choose Boost. Set it to your 48 V battery's charge voltage
`buck_voltage = 13.5`	The 12 V target used when you choose Buck. 13.5 V charges the chassis batteries properly (it's also the D2's maximum)
`buck_current_limit` , `boost_current_limit`	The output current used for Buck (D2 max 178 A) and Boost (D2 max 57 A). Keep them within your wiring and fuses. **Boost is counted on the 48 V side.** The 12 V cable then carries up to about 5 times as many amps. For example, 30 A of boost can mean about 154 A on the 12 V cable. See *Limiting current* in README.md. The maximums are preset for a stock LTV1 (2 AWG, 150 A fuse): 120 A buck, 23 A boost
`..._min` / `..._max` lines	Safety limits. They're preset to the D2 manual's ranges; tighten them if your batteries, cables or fuses need it
`custom_name = DC-DC D2`	The name shown on the Cerbo and in VRM
`password =`	Optional: a password for the web page

Nobody can set a value outside the limits, whether from the Cerbo screen, VRM or the web page.

5e **Type `3` : watch the live readings** (optional, but a good check). The voltages should match what a meter shows on the D2's terminals. To stop, press Ctrl+C, then type N and press Enter.

```
--- can1 10:42:07  ONLINE  0x066=20/s 0x088=100/s 0x520=10/s
LV  13.4 V  +22 A | HV  51.2 V   +6 A | 35 C
Mode: Autonomous (default set-point) | Precharge not in process
Alarms: none
Software 1.2.3  Hardware 5
```

5f **Type `4` : install.** After about 10 seconds you should see:

```
/service/dbus-nomadic-dcdc: up (pid 1834) 6 seconds
... INFO: Nomadic DC-DC driver 1.5.0 starting on auto
... INFO: Converter online on can1
```

Done! It now starts by itself after every reboot and firmware update, in Automatic with limits.

No network to the Cerbo? Install from a USB stick instead

Venus OS can install add-ons from a USB stick or microSD card by itself, with no SSH or password. Do this instead of 5c–5f:

U1 **Get the file.** Download `venus-data-nomadic-d2.tar.gz` from the latest release (https://github.com/CKOCI13/CKOCI13-cerbo-nomadic-d2/releases/latest). It has the standard settings, with `interface = auto`, so there's nothing to edit. To change settings first, edit config.ini (Cerbo Setup option `1`) and type `U` to make your own copy of the file. On a Mac, download it with Chrome or Firefox: Safari can unpack it into a `.tar` file, which the Cerbo ignores.

U2 Copy it to the **top folder** of a USB stick or SD card (FAT32), not inside another folder. Put only **one** `venus-data` file on the stick. If your browser saved it as `venus-data-nomadic-d2 (1).tar.gz`, rename it back to `venus-data-nomadic-d2.tar.gz`: the Cerbo skips names with spaces.

U3 Put the stick in the Cerbo and restart it (**Settings › General › Reboot**, or switch it off and on). The Cerbo unpacks the file and installs the driver while it starts, in Automatic.

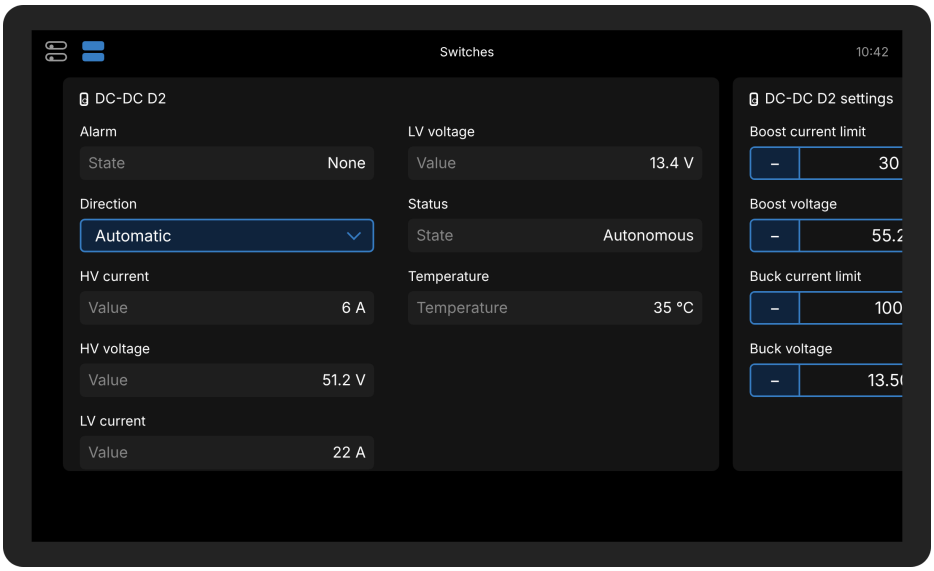
U4 **Wait about 2 minutes, then take the stick out.** The Cerbo renames the file on the stick to `...tar.gz.installed`, so it never installs twice. Then open the switch pane: the DC-DC D2 cards should be there.

> Without SSH you can't run "find the converter" (5c). If the Status card says **No communication**, check the CAN port you chose in 3c, its speed (250 / 500 kbit/s) and the wiring, and restart the Cerbo. To install again, use a fresh file (U1).

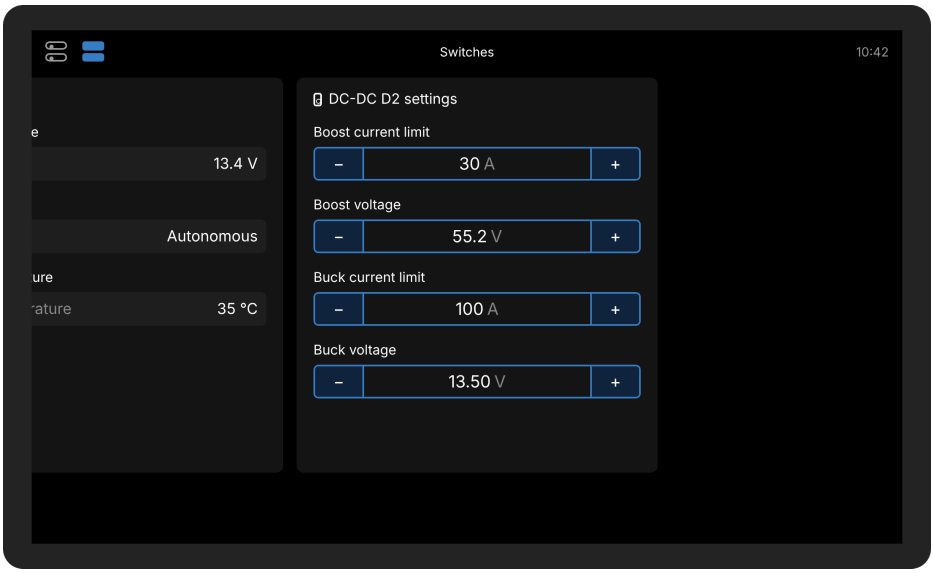
6 Use it

On the Cerbo screen (and Remote Console)

Tap the **switches icon** at the top left of the screen. Two cards appear. Swipe left to see the second one.



DC-DC D2: live readings, the alarm and status, and the Direction dropdown.



DC-DC D2 settings: tap − / + to change a value. It's used whenever the Cerbo is in control (Automatic with limits, Buck or Boost).

Direction: who's in charge

Choice	What happens
Automatic (default)	The Cerbo makes the same choice the D2 would make by itself, and sends your current limits with it. It bucks (charges the 12 V side from 48 V, to about 13.4 V) when the 12 V side drops below 12.5 V, as long as the 48 V side is 41–58 V. It boosts (pushes 12 V energy up to 48 V) when the 12 V side is 13.5–16 V, until it falls below 12.3 V. Between 12.5 and 13.5 V it waits in Standby. Each change waits 2 seconds, and boost pauses for a minute after the 12 V side sags. The web page shows what it chose and why. (With `automatic = hands_off` the Cerbo sends nothing and the D2 decides alone, at its factory limits.)
Buck 48V to 12V	The Cerbo makes the D2 charge the 12 V side, at **Buck voltage** (13.5 V) and up to **Buck current limit**. In Automatic the Cerbo uses about 13.4 V, the D2's own buck target.
Boost 12V to 48V	The Cerbo makes the D2 push energy to the 48 V side, at **Boost voltage** and up to **Boost current limit**.
Off	The Cerbo tells the D2 not to convert.

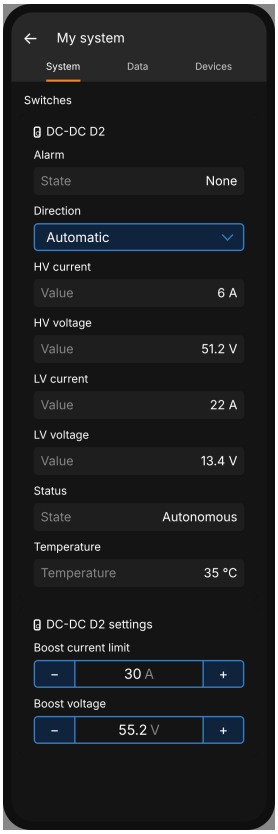
Changing between Buck and Boost always passes through Off for 1 second. Choosing Automatic again lets the Cerbo choose by itself again. The **Alarm** tile shows the most serious active alarm (red Fault label on a fault), and **Status** shows what the D2 reports it's doing, or **No communication**. The precharge controls are hidden because the D2 manual doesn't use precharge. To show them, or to rename or hide any item, go to **Settings › Devices › DC-DC D2**.

> **Limiting current to protect fuses and cables.** The Cerbo's current limits (in Automatic, Buck or Boost) keep the D2 below what a fuse or cable can take, but only while the Cerbo keeps sending its command every 10 ms. The commands stop while the Cerbo restarts or updates, if the driver stops, or if the CAN cable comes loose. Nobody knows yet what the D2 does then. It may go back to full current (up to 178 A into the 12 V side). So: never fit a bigger fuse than the cable is rated for. Size the fuses and cables for the D2's maximum, or ask Nomadic to lower its own limit. Only rely on the Cerbo's limit after the hand-back test below. The project README explains the settings.

In the VRM app and portal

The same cards appear in the **Switches** widget in the VRM app, and in the **Switches** panel on the VRM portal dashboard. The Cerbo must be online with VRM. To change values from VRM, do step 3e (VRM Portal access level: Full).

The picture is an illustration. VRM draws its own version of these cards, so the layout may differ a little.



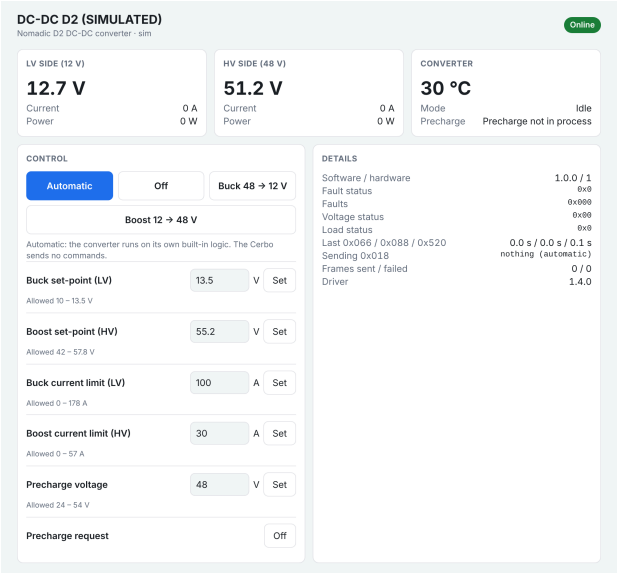
VRM app, Switches widget (illustration)

On a web page

On any phone, tablet or computer on the same network, open:

http://<Cerbo IP>:8088

This page shows everything, including the raw status bits and the exact command being sent. Set a password for it (`password =` in step 5d). Without one, **anyone on your network can operate the converter**.



The web page.

Test the hand-back before relying on Off, Buck or Boost

The Safe Test Plan (`docs/safe-test-plan.pdf`) has the full careful version. In short: it isn't known yet whether the D2 goes back to its own logic when the Cerbo stops sending commands, or keeps following the last command. Do this test once, with the 48 V side between 41 and 58 V, and with `automatic = hands_off` set (option 1, then 4), so that choosing Automatic stops the commands. Set it back to `limited` afterwards:

T1 Choose **Buck**. Status should change to **Buck 48V to 12V**.

T2 Choose **Automatic**. Within a few seconds Status should show what the D2 does on its own: **Autonomous**, **Standby** (12 V side 12.5–13.5 V), **Buck** (below 12.5 V) or **Boost** (13.5–16 V).

T3 Choose **Buck** again, then unplug the CAN cable at the Cerbo. With a meter on the D2's 12 V terminals, watch whether the D2 keeps regulating to 13.5 V, stops, or goes back to its own logic (about 13.4 V). Plug the cable back in.

> The D2 might seem stuck on the last command. For example, it stays in Standby in Automatic while the 12 V side is below 12.5 V. If so, restart it: open the fuses or breakers on **both** LV+ and HV+ (it has no KL15), wait a few seconds, and close them. Then tell Nomadic what you saw. Until you know how it behaves, don't count on the Cerbo's limits while it restarts or updates.

? Troubleshooting

What you see	What to do
Option 2: "not heard on any port"	Check, in this order: 1. the D2 has power: its LV+ terminal (9–16 V) or HV+ terminal (41–58 V), measured to GND. With neither, its controller is off and silent 2. the CAN speed: if it reports bus errors, switch to the other speed (step 3c) 3. the wires (2b), and that each lever connector holds the right D2 wire (CAN_H with white/brown, CAN_L with brown) 4. the termination reads about 60 Ω (2h) 5. the cable is in the Cerbo port you set up
"ssh" or "tar" is not recognized	Install the Windows OpenSSH client: Settings › Apps › Optional features › Add a feature › OpenSSH Client.
"Connection timed out" or "Connection refused"	Check the IP address (menu option 6), that the PC and Cerbo are on the same network, and that Enable SSH on LAN is on (step 3d).
"Permission denied"	Wrong root password. Set it again (step 3d, item 3).
"REMOTE HOST IDENTIFICATION HAS CHANGED"	This happens after the Cerbo is reset or replaced. Use menu option 7, then try again.
Option 2: Venus OS version warning	Update the Cerbo (step 3a). Older firmware can't show the cards.
Nothing appears in the switch pane	Check the firmware is v3.80 or newer. Use option 5 (log) and look for "Registered ... on D-Bus". Make sure `switch_pane = yes` in config.ini.
Status shows Standby in Automatic	Normal when the 12 V side is between 12.5 and 13.5 V: the D2 has nothing to do.
Buck doesn't start	The D2 only bucks with the 48 V side between 41 and 58 V (D2 manual), even when the Cerbo asks.
Status shows No communication	The Cerbo can't hear the D2. Check that its LV+ or HV+ terminal is live, the cable and the termination.
The log says "CAN send failed"	Only happens when the Cerbo is sending commands (Automatic with limits, Off, Buck or Boost). Nothing on the bus is acknowledging: same checks as above. The speed is the usual cause.
A number won't go higher or lower	It has reached a limit from config.ini. To change the limit: option 1, edit, save, then option 4.
VRM shows the cards but won't accept changes	Set VRM Portal access level to Full (step 3e). The Cerbo must be online.
The "DC system" figure on the Cerbo looks wrong	The Cerbo counts the D2 as a DC-DC charger. Set `service_class = nomadicdcdc` in config.ini (option 1, then 4). The D2 leaves the Devices list, but the cards, VRM and web page keep working.

To change settings later, open Cerbo Setup, choose 1 to edit config.ini, save, then choose 4. The driver restarts with the new settings. Values you change on the screen, in VRM or on the web page are remembered automatically. **To remove it,** choose 8 and type YES. The D2 carries on in its own automatic mode.

> Before you rely on the Cerbo's limits, do the hand-back test in Step 6. While it's in control (Automatic with limits, Buck, Boost or Off) the Cerbo sends a command every 10 ms. If the Cerbo restarts or the cable comes loose, those commands stop, and it isn't yet known whether the D2 then goes back to its own logic.

Points to confirm with Nomadic

- What the D2 does when CAN commands stop: does it go back to its own logic, and after how long? (Your hand-back test will show this too)
- Whether it sends its status even when it isn't receiving commands (the driver assumes it does)
- Reprogramming the factory limits and buck target to suit your wiring, so they also hold while the Cerbo restarts
- Which direction counts as positive current in the readings